



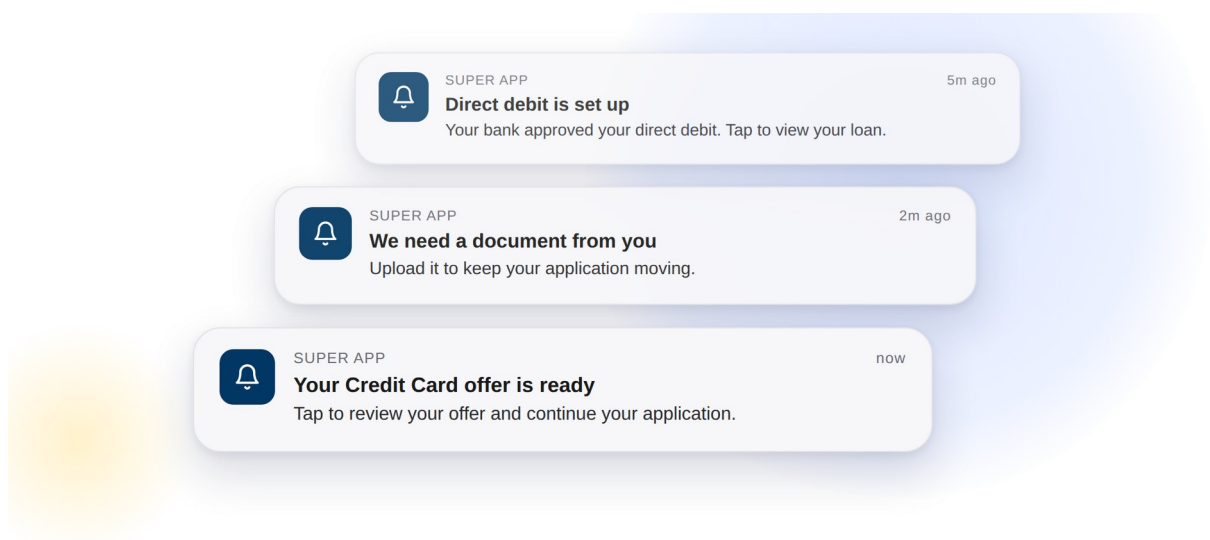
BUSINESS REQUIREMENTS DOCUMENT

Push Notification

Customer journey alerts in the Super App — Credit Card, Personal Loan and CASA

V1.0 / 28 September 2026

Prepared for Reem Bank



Appro Onboarding Solutions FZ-LLC · Customer Journey

This is not a legally binding document. Highly confidential not to be shared without written consent.

Based on the Reem Bank middleware Push Notifications specification v0.1 (Avanza, 25 September 2026). Items awaiting confirmation are listed in section 10.

VERSION HISTORY

Date	Version	Author	Change Description
28-09-2026	1.0	Hailey, Appro	Created BRD

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1. FEATURE OVERVIEW AND KEY COVERED AREAS

This document describes the requirements for **Push Notification** in the Reem Bank customer journey for **Credit Card, Personal Loan and CASA**, on iOS and Android.

Today the customer learns about decisions and required actions on an application — a credit decision, a Direct Debit Authority response, the end of the cooling-off period — by **email and in-app notification only**. When the customer is outside the Super App, nothing reaches the device, and the application waits until the customer returns.

Push Notification adds a device-level alert at each agreed trigger event. The Appro platform sends the push request to the Reem Bank middleware Push API, following the same integration pattern as the existing SMS and email integration. When the customer taps the notification, the Appro journey opens at the screen where the application continues.

This BRD covers the following areas:

- **Trigger events and push content** — eleven notifications, each with its title, body and the screen it opens
- **Language** — English or Arabic, following the customer's current language
- **Middleware integration** — the Push API request, headers and response
- **Response handling and automatic retry**
- **Duplicate and reminder control**
- **Audit trail** — one new step per push
- **Business rules** — personal data and excluded templates

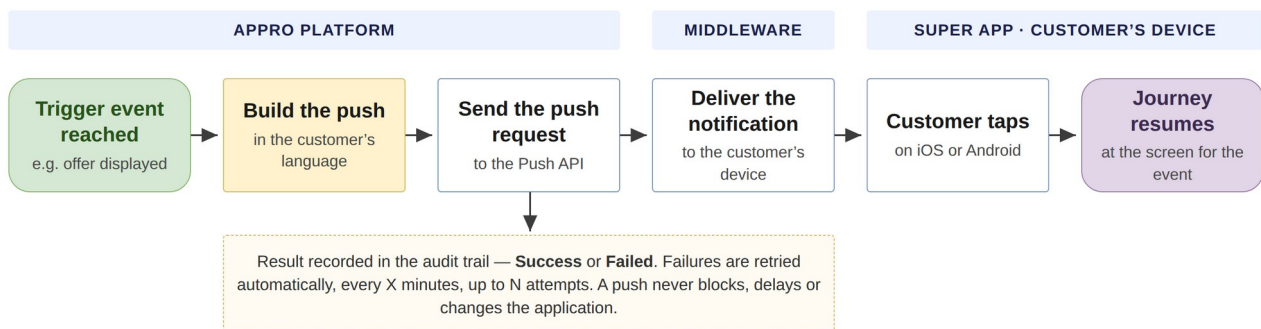


Figure 1 — How a push reaches the customer (green = start, amber = system action, blue = step, purple = end)

Key steps

1. A trigger event is reached on the application, for example an offer is displayed or a document is requested.
2. The system builds one push for the event: the Push ID, the title and the body, in the customer's current language.
3. The system sends the push request to the Reem Bank middleware Push API.
4. The middleware delivers the notification to the customer's device through the Super App.
5. The customer taps the notification, and the Appro journey opens at the screen for that event.
6. The result is recorded in the application audit trail. A failed push is retried automatically and never blocks the application.

Note: Push content is fixed per notification and is not configurable in Super Portal in this release. Customer actions on a notification (open, dismiss, ignore) are not tracked. Existing email and in-app notifications are unchanged.

2. SCOPE

2.1 Product scope

Credit Card, Personal Loan and CASA, on **iOS and Android**. Each notification applies only to the products listed against it in section 3.

2.2 In-scope notifications

- **P01** — Offer ready (Credit Card, Personal Loan)
- **P01 (CASA)** — Account approved in principle (CASA)
- **P02** — Offer expiry reminder, once per day for 15 days (all products)
- **P03** — Offer expired (all products)
- **P04** — Update on the application after a rejection (all products)
- **P05** — Document requested from the customer (all products)
- **P06** — Application active again after the cooling-off period (Credit Card, Personal Loan)
- **P08** — Direct debit set up (Personal Loan)
- **P09** — Direct debit rejected by the customer's bank (Personal Loan)
- **P10** — Loan on its way after disbursement (Personal Loan)
- **P11** — Product ready: card issued (Credit Card, CASA) or loan approved (Personal Loan)

2.3 Not in scope

- OTP messages — never sent as push (BR3).
- Bank and internal-user notifications — never sent as push (BR4).
- Configuring push content in Super Portal — content is fixed per notification in this release.
- Tracking customer actions on a notification — open, dismiss or ignore.
- Changes to the existing email and in-app notifications — they remain unchanged (BR1).

3. TRIGGER POINTS AND PUSH CONTENT

3.1 Trigger events

A push is fired at the same trigger point as the existing email and in-app notification templates listed under Source templates. Existing email and in-app notifications remain unchanged.

Push ID	Priority	Applies to	Trigger event	Source templates
P01	P0	CC · PL	AIP given (at least one offer displayed), or Credit Queue approves / overrides the application	ET15/NT06 · ET9
P01 (CASA)	P0	CASA	Account application approved in principle	ET15/NT06 (CASA)
P02	P0	CC · PL · CASA	Scheduled reminder once per day for 15 days, starting after the first offer is displayed	ET13 · NT02
P05	P0	CC · PL · CASA	Credit or Sales team returns the case requesting documents from the customer	NT12
P06	P0	CC · PL	Cooling-off end date reached (start + 5 days, configurable) with no customer action; application returns to Approved In Principle	ET55/NT24
P09	P0	PL	Customer's bank rejects the Direct Debit Authority	ET41/NT17
P04	P1	CC · PL · CASA	Queue user rejects (Credit, Sale, Risk or Compliance), Rule Engine filtration fails, or minimum income not met	ET22/NT10 · ET8/NT11 · ET12 · ET19/NT07/NT08 · ET21/NT19
P08	P1	PL	Customer's bank accepts the Direct Debit Authority	ET40/NT21
P10	P1	PL	Disbursement Checker approves and releases the disbursement	ET42/NT18
P11	P1	CC · PL · CASA	Card issued (CC, CASA) or loan approved (PL)	ET37 · ET38 · NT13
P03	P2	CC · PL · CASA	Offer validity (30 days) elapses with no acceptance	ET14/NT03

Source templates: **ET** = email template, **NT** = in-app notification template, as configured in Communication Setup. Priority: P0 highest.

3.2 Push content

Push ID	Title (EN, max 40 characters)	Body (EN, max 100 characters)	Opens at (deep-link screen)
P01	Your %%PRODUCT_TYPE%% offer is ready	Tap to review your offer and continue your application.	Offer selection
P01 (CASA)	Your account is approved	Tap to continue and complete your account opening.	Resume application
P02	Your offer expires in %%COUNTING_DOWN%% days	Complete your application before your approval expires.	Offer selection

Push ID	Title (EN, max 40 characters)	Body (EN, max 100 characters)	Opens at (deep-link screen)
P05	We need a document from you	Upload it to keep your application moving.	Document upload
P06	Your application is active again	Your cooling-off period has ended. Tap to continue or cancel.	Application Ready to Continue screen, then KFS
P09	Action needed on your direct debit	Your bank didn't accept the authority. Tap to see next steps.	DDA screen / next steps
P04	Update on your application	Tap to view the status of your %%PRODUCT_TYPE%% application.	Application status
P08	Direct debit is set up	Your bank approved your direct debit. Tap to view your loan.	Loan details
P10	Your loan is on its way	Funds will reach your account shortly. Tap for details.	Loan details
P11	Your %%PRODUCT_TYPE%% is ready	Tap to see what happens next.	Product screen (card / loan / account)
P03	Your offer has expired	You can reapply whenever you're ready.	SDK start (new application)

SUPER APP

Your Credit Card offer is ready

Tap to review your offer and continue your application.

now

P01 CC · PL → opens **Offer selection**

SUPER APP

Your account is approved

Tap to continue and complete your account opening.

now

P01 (CASA) CASA → opens **Resume application**

SUPER APP

Your offer expires in 14 days

Complete your application before your approval expires.

now

P02 CC · PL · CASA → opens **Offer selection**

SUPER APP

We need a document from you

Upload it to keep your application moving.

now

P05 CC · PL · CASA → opens **Document upload**

SUPER APP

Your application is active again

Your cooling-off period has ended. Tap to continue or cancel.

now

P06 CC · PL → opens **Application Ready to Continue, then KFS**

SUPER APP

Action needed on your direct debit

Your bank didn't accept the authority. Tap to see next steps.

now

P09 PL → opens **DDA screen / next steps**

SUPER APP

Update on your application

Tap to view the status of your Personal Loan application.

now

P04 CC · PL · CASA → opens **Application status**

SUPER APP

Direct debit is set up

Your bank approved your direct debit. Tap to view your loan.

now

P08 PL → opens **Loan details**

SUPER APP

Your loan is on its way

Funds will reach your account shortly. Tap for details.

now

P10 PL → opens **Loan details**

SUPER APP

Your Credit Card is ready

Tap to see what happens next.

now

P11 CC · PL · CASA → opens **Product screen (card / loan / account)**

SUPER APP

Your offer has expired

You can reapply whenever you're ready.

now

P03 CC · PL · CASA → opens **SDK start (new application)**

Illustrative. English copy as specified; %%PRODUCT_TYPE%% and %%COUNTING_DOWN%% shown with example values. The final look depends on the device operating system and the Super App. Arabic copy is to be provided by Reem Bank Business.

Figure 2 — The eleven notifications as the customer sees them (illustrative)

3.3 Content rules

- When the application reaches a trigger event and the product matches Applies to, the system must build **one push request** for that application.
- [%%PRODUCT_TYPE%%] and [%%COUNTING_DOWN%%] must be replaced with actual values before sending. No placeholder is ever shown to the customer.
- The title must not exceed **40 characters** and the body **100 characters**, after replacement.
- Content is fixed per Push ID and is not configurable in Super Portal in this release.

3.4 Language

- The push is sent in **English or Arabic**, following the language the customer is currently using.
- **⚠ TBC by Reem Bank Business:** Arabic title and body for every notification.

4. PUSH NOTIFICATION API INTEGRATION

The integration follows the Reem Bank middleware specification **Reem Payments API — Push Notifications v0.1** (Avanza, 25-09-2026). Items the specification does not yet answer are flagged **⚠ TBC by Avanza** and tracked in section 10.

4.1 Request details

Item	Value
Method	POST
Endpoint	http://<MW URL>/api/notifications/push ⚠ TBC by Avanza: MW URL per environment (SIT, UAT, PROD) and HTTPS
Authentication	Bearer token ⚠ TBC by Avanza: how the token is issued (endpoint, grant, expiry)

4.2 Header parameters

Parameter	Type	Required? (Y/N)	Description	Values / Data Source
stan	String	Y	Unique system audit trace number in each request	Generated by Appro, unique per request. ⚠ TBC by Avanza: format and length
Authorization	String	Y	Authorization header	Bearer <token>
Content-Type	String	Y	Request format	application/json
channel_id	String	Y	Channel id of the calling system	⚠ TBC by Avanza: value assigned to Appro

4.3 Request parameters

Parameter	Type	Required? (Y/N)	Description	Values / Data Source
requestId	String	Y	Unique message id	Generated by Appro, unique per push. ⚠ TBC by Avanza: same or new id on retry
timestamp	String	Y	Request date-time	ISO 8601 UTC with milliseconds, e.g. 2026-09-28T06:49:02.366Z

Parameter	Type	Required? (Y/N)	Description	Values / Data Source
type	String	Y	Notification type	Fixed: APPRO_NOTIFICATION
action.id	String	Y	Push identifier	Push ID from section 3, e.g. P01. P01 (CASA) is sent as P01 with action.value = CASA
action.type	String	Y	NONE / POPUP / APPRO_JOURNEY	APPRO_JOURNEY for every Push ID, so a tap opens the Appro journey. ⚠ TBC by Avanza: behaviour of NONE / POPUP, and how the Super App hands action.id and action.value to the Appro journey
action.value	String	Y	Product	CREDIT_CARD / PERSONAL_LOAN / CASA, from the application. NA is not used — every push belongs to an application
recipient.mobilePhone	String	Y	Customer mobile number	From the application. ⚠ TBC by Avanza: format (country code, '+', spaces)
recipient.customerId	String	Conditional	Customer CIF — required if ETB	ETB: customer CIF. NTB: not sent. ⚠ TBC by Avanza: whether mobilePhone alone reaches an NTB customer
recipient.email	String	N	Customer email	Not sent. ⚠ TBC by Avanza: purpose
content.title	String	Y	Push title	Per section 3.2, in the customer's current language
content.body	String	Y	Push body	Per section 3.2, in the customer's current language
content.message	String	N	Notification message	Sent empty (""). ⚠ TBC by Avanza: purpose versus body
content.language	String	Y	Language — en / ar	Customer's current language. ⚠ TBC by Avanza: case (en vs EN in the sample) and type (listed as Date Timestamp)

4.4 Response parameters

Parameter	Type	Description	Values / Handling
messageId	String	Message id returned by the middleware	Recorded in the audit trail
error.code	String	Result code	"000" = Processed Ok ⚠ TBC by Avanza: full code list, and which codes are retryable
error.message	String	Result description	Recorded in the audit trail with error.code

A sample request and response are in Appendix 3.

5. RESPONSE HANDLING AND RETRY

- If the HTTP code is **20x** and [error.code] = **"000"** (Processed Ok) → the push is sent → audit trail Step Status = **Success** → the journey proceeds.
- Else → the system retries automatically after **X minutes**, up to **N attempts**. X and N are configurable.
 - "Else" covers a timeout, any non-20x HTTP code and any [error.code] other than "000".
 - **⚠ TBC by Avanza:** the full error-code list, and any code that should not be retried.
- If the push is still not successful after the last attempt → audit trail Step Status = **Failed** → the journey proceeds. A push failure must not block, delay or change the application status.
- The system tracks the API delivery result only. Customer actions on the notification (open, dismiss, ignore) are not tracked.

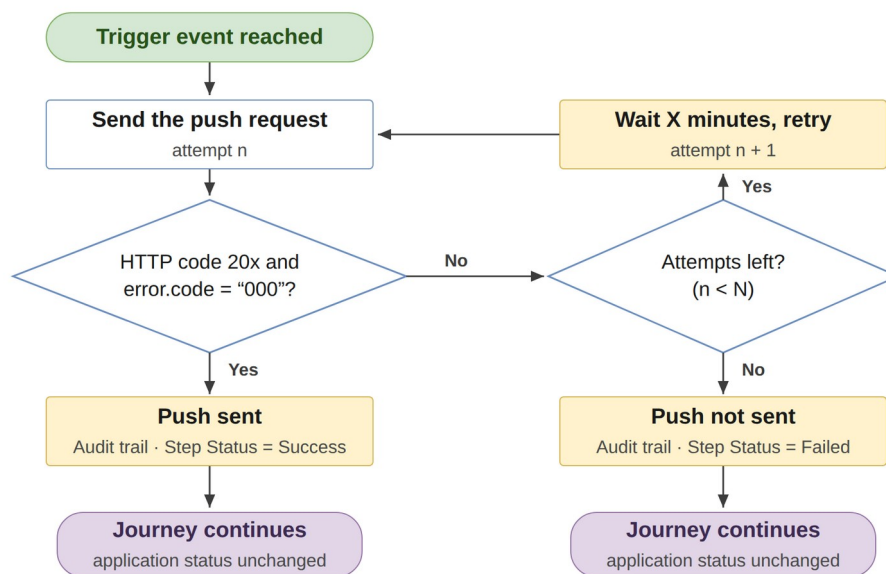


Figure 3 — Response handling and automatic retry (X and N configurable)

6. DUPLICATE AND REMINDER CONTROL

- The system must send at most **one push per Push ID per trigger occurrence per application**. Consolidated templates (for example P04) must not generate more than one push for the same decision.
- **P02** is sent **once per day for 15 days** from the first offer display, and stops immediately once the customer selects an offer or the application is expired, cancelled or rejected.

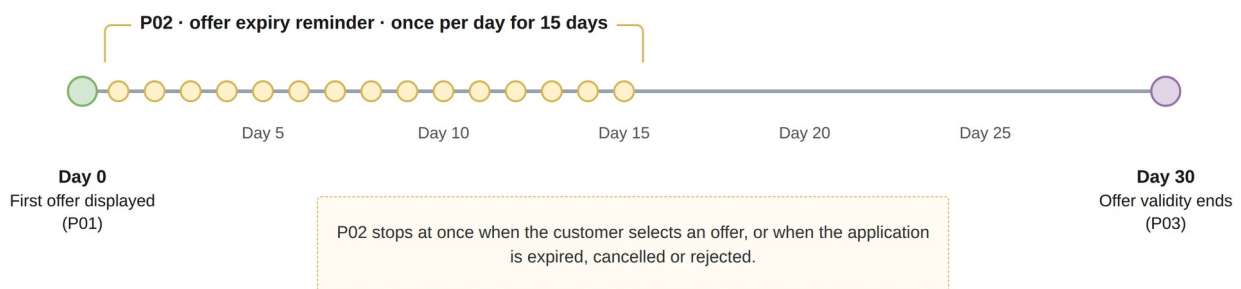


Figure 4 — The P02 offer expiry reminder

7. AUDIT TRAIL

Every push writes one entry to the application audit trail, visible in Application Enquiry. The application status is not changed.

Field	Value
[Application ID]	Current Application ID
[Step]	Push Notification
[Step Details]	Send Push <Push ID> · stan · requestId · messageId — or error.code and error.message when the push is not sent
[Start Time]	Date-time the request is sent
[End Time]	Date-time of the response, or of the last timeout
[Attempt No.]	n of N
[Step Status]	Success / Failed
[Action by]	System

The Audit Trail screen has no attempt column, so the attempt is shown in Step Details, as in Figure 5.

No	Step	Start Time	End Time	Step Details	Current Application Status	Step Status	Action By
27	Push Notification	28/09/2026 10:15:02	28/09/2026 10:15:03	Send Push P01 · Attempt 1 of 3 stan 000187 · requestId 8F3A92C1 messageId 0:1790341509514173...	Approved In Principle	Success	System
28	Push Notification	29/09/2026 10:00:00	29/09/2026 10:10:30	Send Push P02 · Attempt 3 of 3 stan 000214 · requestId 5C71D0B4 No response (timeout)	Approved In Principle	Failed	System

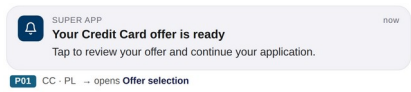
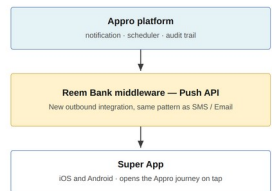
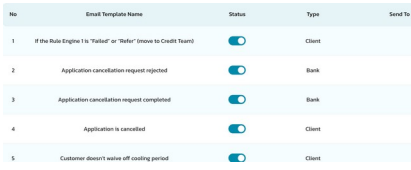
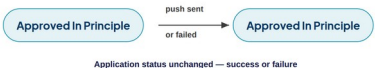
New step: Push Notification — one entry per push (illustrative values)

Figure 5 — Application Enquiry > Audit Trail — the new Push Notification step (composite over the UAT screen, illustrative values)

8. BUSINESS RULES

No.	Rule
BR1	Push is additional; existing email and in-app notifications are unchanged.
BR2	No personal data in the title or body: no name, Application ID, IBAN or amounts.
BR3	OTP content (ET17, ET18, ET35, NT04, NT05) is never sent as push.
BR4	Bank and internal-user templates are never sent as push.

9. IMPACT ANALYSIS

Area	Impact	Screen
Customer journey (Super App)	A device notification at eleven trigger events. A tap opens the Appro journey at the screen for that event; existing journey screens are reused as the destinations.	
Middleware integration	New outbound call from the Appro platform to the Reem Bank middleware Push API, following the SMS and email integration pattern. Endpoint, credentials and channel id per environment: see section 10.	
Communication Setup	No change. Existing email and in-app templates keep working as today and remain the trigger reference for each push. Push content is fixed per notification in this release and is not added to Communication Setup.	
Audit trail	One new step, Push Notification, per push — with Success or Failed, the attempt count and the middleware references for tracing.	
Application status	No change. A push — sent or not — never changes, blocks or delays the application.	
Scheduling	A daily run sends the P02 reminder for 15 days and stops as soon as the offer is selected or the application is expired, cancelled or rejected.	
Reporting	The API delivery result is tracked. Customer actions on a notification (open, dismiss, ignore) are not tracked.	No screen change

10. OPEN QUESTIONS

Each item carries Appro's proposal. Confirming the proposal is enough to proceed.

#	Question	Owner	Appro proposal
1	MW URL for SIT, UAT and PROD. The specification shows http:// — confirm the endpoint is served over HTTPS.	Avanza	HTTPS in all environments
2	How Appro obtains the Bearer token: token endpoint, grant type, expiry and refresh; UAT credentials.	Avanza	Please advise
3	stan: format, length and uniqueness scope.	Avanza	Generated by Appro, unique per request, including each retry
4	channel_id value assigned to Appro, per environment.	Avanza	Please assign
5	requestId: format and length; on a retry, resend the same requestId so the middleware can de-duplicate?	Avanza	Same requestId on every retry of the same push
6	action.type: behaviour of NONE, POPUP and APPRO_JOURNEY; how action.id and action.value reach the Appro journey on tap; when NA applies.	Avanza	APPRO_JOURNEY for every push; NA not used
7	recipient.mobilePhone format: country code, '+', spaces (the sample value has a trailing space).	Avanza	971XXXXXXXXX — country code, digits only
8	NTB customers have no CIF: is mobilePhone alone enough to reach them; omit customerId or send it empty?	Avanza	Omit customerId for NTB
9	Purpose of recipient.email and of content.message versus body.	Avanza	email not sent; message sent empty
10	Maximum title and body length supported by the Super App.	Avanza	Title 40, body 100 characters
11	content.language type (listed as Date Timestamp) and case (en / EN); timestamp format and time zone.	Avanza	String, lower case en / ar; ISO 8601 UTC with milliseconds
12	Full error.code list and messages, the HTTP codes returned, and any code that should not be retried.	Avanza	Success = HTTP 20x with "000"; anything else retried up to N
13	Does "000 Processed Ok" mean accepted for delivery, or delivered to the device?	Avanza	Recorded as sent, not delivered
14	Recommended request timeout and any rate limit (P02 runs daily).	Avanza	Please advise
15	Arabic title and body for every notification.	Reem Bank Business	To be provided

APPENDIX 1: SUMMARY OF CHANGES

Areas	Previous Behaviour	New Behaviour
Customer alerts	Email and in-app notification only	Email, in-app notification and push notification
Customer outside the Super App	No device-level alert; the application waits for the customer to return	A device notification at eleven trigger events
Resuming the journey	The customer opens the Super App and navigates to the application	A tap opens the Appro journey at the screen for the event
Offer expiry reminder	Email and in-app reminders (ET13, NT02)	Plus a daily push (P02) for 15 days, stopping on selection, expiry, cancellation or rejection
Audit trail	No push entries	One Push Notification entry per push, with the result and the middleware references

APPENDIX 2: WORKFLOW

1. The application reaches a trigger event listed in section 3.1, and the product matches Applies to.
2. The system builds one push: Push ID, title and body in the customer's current language, with placeholders replaced.
3. Duplicate control applies: at most one push per Push ID per trigger occurrence per application.
4. The system sends the request to the middleware Push API (attempt 1).
5. If the HTTP code is 20x and [error.code] = "000" → the push is sent; the audit trail records Success.
6. Otherwise the system waits X minutes and retries, up to N attempts.
7. After the last unsuccessful attempt, the audit trail records Failed.
8. In every case the journey continues and the application status is unchanged.
9. The middleware delivers the notification; on tap, the Super App opens the Appro journey at the deep-link screen.

The end-to-end flow is shown in Figure 1 and the retry logic in Figure 3.

APPENDIX 3: API REFERENCE

Sample request

```
{
  "requestId": "<unique id>",
  "timestamp": "2026-09-28T06:49:02.366Z",
  "type": "APPRO_NOTIFICATION",
  "action": { "id": "P01", "type": "APPRO_JOURNEY", "value": "CREDIT_CARD" },
  "recipient": { "mobilePhone": "9715XXXXXXX", "customerId": "<CIF, if ETB>" },
  "content": {
    "title": "Your Credit Card offer is ready",
    "body": "Tap to review your offer and continue your application.",
    "message": "",
    "language": "en"
  }
}
```

Sample response

```
{
  "messageId": "0:1790341509514173%d5881a7ad5881a7a",
  "error": { "code": "000", "message": "Processed 0k" }
}
```

Product mapping

Product on the application	action.value
Credit Card	CREDIT_CARD
Personal Loan	PERSONAL_LOAN
CASA	CASA

APPENDIX 4: GLOSSARY

Term	Meaning
AIP	Approval In Principle — at least one offer is displayed to the customer
Appro journey	The Appro onboarding journey (SDK) that runs inside the Super App
CIF	The bank's customer identification number
DDA	Direct Debit Authority
ETB / NTB	Existing-to-Bank / New-to-Bank customer
KFS	Key Fact Statement
Middleware (MW)	The Reem Bank integration layer, provided by Avanza
Push ID	The identifier of each notification, P01 to P11
stan	System audit trace number, unique per request
Super App	The Reem Bank mobile app on iOS and Android

THANK YOU

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Customer Journey

